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Growth of indeterminate choroidal melanocytic tumors: Time to malignant transformation

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Abstract:

Indeterminate choroidal melanocytic tumors may exhibit growth under observation, but the clinical significance of early versus late growth remains unclear. In this review we aim to explore the temporal profile of growth to detect benign and malignant subsets. There was no specific set of tumor dimensions or characteristics applied for exclusion or inclusion except that all included tumors had to be described as indeterminate or suspicious nevus and observed for growth by the authors. Nine published studies (1977–2021) that included 8766 patients were reviewed. The primary outcomes were time to event (growth) and its correlation with the growth rate of small choroidal melanoma. The studies were largely retrospective and had differing inclusion criteria: largest basal diameter of 7.5–24 mm (mean, 14.83 mm) and maximum height of 2–6.7 mm (mean, 4.19 mm). Most defined growth by a change in height of >0.3 mm or base of >0.5. Among 8606 tumors followed, 478 (5.6%) showed growth over 5 years, with growth rates in individual studies ranging from 2% to 55% (mean: 19.0%). The highest event rate occurred in the 1st year (range: 0.7%–15%; mean: 6.5%), stabilizing to 0%–5% (mean: 1.5%) by year 5. The majority (range: 52%–83%; mean: 67%) were observed to grow in the initial 2 years (early growth). Over subsequent years (late growth), the proportion of growing tumors reduced and stabilized by years 4 and 5 to a mean of 6.1% and 8.3%, respectively. Time-to-event analysis indicates the heterogeneity of IMTs. Tumors growing in the first 2 years (early growth) may be melanoma in the evolution and their characteristics previously identified as the risk factors for growth may be interpreted as the factors predictive or diagnostic of small choroidal melanoma. In contrast, tumors demonstrating growth in subsequent years (late growth), after documented stability, may represent malignant transformation of a choroidal nevus.

Keywords:

Choroidal melanoma, choroidal nevus, indeterminate melanocytic lesion, indeterminate melanocytic tumor, small choroidal melanoma

Introduction

The description of an “indeterminate pigmented choroidal tumor” introduced by Butler *et al.* in 1994^[1] is applied to melanocytic lesions in the choroid that are approximately 2.5 mm or less in height and 10 mm or less in their largest basal diameter (LBD). Over the years, expansion of the term to include tumors (5.0–16.0 mm in) and 1.0 mm to 2.5 mm in height) corresponding to small choroidal melanoma

in the Collaborative Ocular Melanoma Study (COMS) reports, seems appropriate as a majority (about two-thirds) remained stable without treatment, revealing them to be nevi rather than melanoma.^[2] The questions that surround these lesions have been discussed since 1977 by Gass^[3] reflecting enduring concern as reviewed in a recent editorial.^[4]

Several retrospective case series have described methods to identify melanoma within a population of indeterminate melanocytic tumor (IMTs).^[1,3,5-13] Current

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clinical practice revolves around two broad categories of diagnostic decision-making: tumor sampling for cytological/histopathological assessment or observation to document malignant growth.^[14]

Tumor sampling from IMT involves the precarious step of placing a small gauge needle or vitrector in a thin tumor that is adjacent to major retinal vessels as most IMTs are located in proximity to the disc, macula, and the uvea being one of the most blood vessel rich tissues.^[15,16] In addition to the risk of visually significant complications such as subretinal or vitreous hemorrhage, the sample yield may be low with the 68% with fine-needle aspiration biopsy.^[8-11] However, high rates of positivity for prognostication with 27G vitreous cutter can be achieved.^[17]

Observation to document growth and subsequent treatment presumes that tumor growth indicates malignancy.^[12,18] Besides providing an evidentiary basis for diagnosis, such an approach may avoid unnecessary treatment for some substantial number of patients (approximately two-thirds in COMS) who will remain unchanged for years.^[18,19] In addition, IMTs that are assessed presumptively on clinical diagnosis to be choroidal melanoma have a low risk of metastasis (1%–3%);^[20] therefore, detecting growth provides little added risk if a tumor is observed for a short period of time.^[14,21-23] Based on the known growth rate of increase in LBD of 1–2 mm/year in biopsy-proven choroidal melanoma,^[7] growth of the IMT would be observable within 6–12 months.^[2,24-26] Such observable growth over months may be sharply distinguished from minimal growth of choroidal nevi (0.015–0.03 mm/year) detected over decades only by comparison of high-quality photographs.^[27] Others have proposed treatment on the basis of “risk factors” for growth prior to a demonstration of any growth.^[28]

The research question explored in this review is when the growth of an IMT signals a malignant transformation. Several aspects of growth of IMT including definitions of growth, dimensions of growth, growth rates, and metastatic events that occur during observation periods have been studied.^[7,28,29] Upon critical review, analysis of time to growth (time-to-event analysis) may provide insights into the nature of the tumor (benign vs. malignant) and remains unexplored.^[4] In this review, we extract and analyze the data specifically regarding time-to-growth events from published studies and interpret the results in light of known growth rates of small choroidal melanoma.

Methods

The study adhered to the PRISMA protocol for the data search. The review protocol was not required

to be registered in any database such as PROSPERO. A search of MEDLINE, PubMed, Embase, Cochrane Database, Clinical Trials, and Scopus was conducted. Key words included: “indeterminate melanocytic lesion,” “indeterminate melanocytic tumor,” “small choroidal melanoma,” “suspicious nevus,” “nevus growth,” “malignant conversion of nevus,” “malignant transformation choroidal melanoma,” and “suspicious choroidal melanoma.” Additional references were culled from the bibliographies of these references. References spanned the period 1959–2023 and were not exclusive to the English language. References were evaluated for their pertinence to the topic, with attention given to papers detailing clinical outcomes of growth of indeterminate tumors that authors felt had a potential for malignant transformation.

There was no specific set of tumor dimensions or characteristics applied for exclusion or inclusion except that all included tumors had to be described as indeterminate or suspicious nevus and observed for growth by the authors. There was a general presumption that indeterminate tumors would fall somewhere within the small COMS tumor definition (5.0–16.0 mm in LBD and 1.0 mm to 2.5 mm in height).^[2] If a study included tumors above this range, they were included so long as the authors felt that they were indeterminate or had a possibility of being a nevus. Included works had to describe a time frame in which growth was detected that was either directly described or could be calculated using units of time reported by authors. Articles that were repetitious or decidedly nonclinical were omitted from consideration.

Once relevant articles containing needed details were identified, they were studied for tumor dimensions, definitions of growth, methods of growth detection, risk factors for growth, and time to growth. The proportion of cases demonstrating growth per year was calculated for the initial period of 5 years of observation. We created a plot of the data showing cumulative percent with growth on the y-axis according to the year on the x-axis. The individual raw data from all studies were color coded. The locally estimated scatterplot smoothing (LOESS) for these studies is shown in a black dashed smooth line. A graphic comparison was undertaken by overlaying LOESS for the proportion of tumors demonstrating growth from dataset of 42 eyes with biopsy-proven choroidal melanoma that were observed prior to intervention, shown in a black solid smooth line.^[7]

Results

A total of 18,427 articles were found on an initial search. Following exclusion due to repetitious or irrelevant articles or reports that do not list prerequisite details, 9

studies were included for analysis [Supplement Figure 1]. A total of 8606 tumors were observed across all studies with a mean LBD of 14.83 mm (range 7.5–24) and a mean height of 4.19 mm (range 2–6.7) [Table 1].^[1,2,6,25,29-33] Of the 9 studies, all except one^[29] included tumors above 2.5 mm in height and 2 included tumors above 16 mm in base.^[6,33] Greatest agreement is represented as darkest/overlapping areas (LBD 5.0–16.0 mm and height 1.0 mm to 2.5 mm) in Figure 1.

With one exception (COMS, prospective study),^[2] all studies were retrospective.^[1,2,6,25,29-33] Growth was loosely defined in older publications (through the 1990s); however, since the 2000s, authors have used uniform terminology and definitions. For example, 4 of the 9 studies that had no numerical definition of growth were all published before 1994.^[1,25,30,31] Recent studies (4 studies) agreed that >0.5 change in LBD or >0.3 mm change in height constituted growth though the combination of these features was not uniformly shared.^[6,29,32,33] The methodology for documenting growth also changed over time with photography and ultrasonography performed in all included participants since the 1990s [Table 2].^[2,6,29,32,33]

Analysis of risk factors in all 9 studies revealed that the tumor size was directly and positively

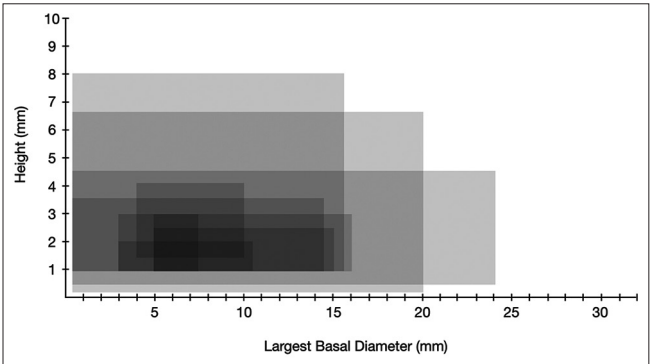


Figure 1: Tumor size graphically represented in grey rectangles. Where there is the greatest overlap, there is also the most agreement as to the size of a tumor that could be described as indeterminate. The darkest areas indicate the areas of most agreement

associated with growth (tumor height in all 9 studies, LBD Tumor in 7 studies).^[1,6,25,29-34] Another risk factor with high correlation across studies was the presence of orange pigment in 8 studies.^[1,6,25,29,31-34] Juxtapapillary location was identified as a risk factor in 4 studies.^[29,31-33] There was nearly uniform agreement (8 of 9 studies) regarding the presence of drusen and RPE degeneration as a negative or neutral association for growth [Table 3].^[1,6,25,29-31,33,34] Only one study found a positive association.^[32]

Of the 8606 tumors studied, 478 were observed to grow during the initial 5-year period of observation resulting in a mean growth rate of 5.6% (range 2%–55%) [Table 1]. An analysis of the timing of growth among tumors that grew revealed that a mean of 39% (range 28%–58%) demonstrated growth in the 1st year. This rate decreased over time to a mean of 28% (range 8%–35%) by year 2. Overall, the majority (range 52%–83%; mean 67%) of tumors that grew did so in the initial 2 years (early growth). Over subsequent years (late growth), 8% (range 13-23%), 6.1%, and 8.3% of tumors that grew did so in years 3, 4, and 5, respectively [Figure 2].

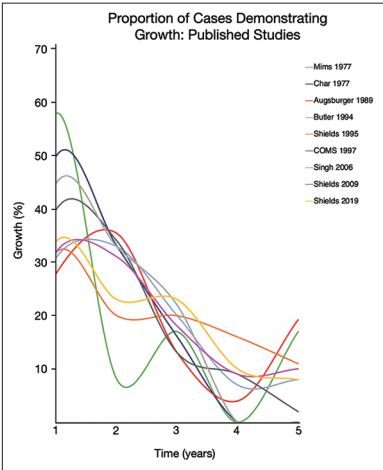


Figure 2: Time-to-event by year among tumors that grew in the observation period (up to 5 years)

Table 1: Indeterminate melanocytic tumors observed for growth

Author, year	Number of tumors	Number of growth	Tumor size				Growth rate (%)					Total (%)
			Base		Height		1	2	3	4	5	
			Minimum	Maximum	Minimum	Maximum						
Mims, 1977	61	12	3	7.5	1	3	11	2	3	0	3	20
Char, 1977	20	11	3	10.5	1	2	15	10	5	0	0	55
Augsburger, 1989	197	50	0.5	15.5	1	8	7	9	3	1	5	25
Butler <i>et al.</i> , 1994	293	98	4	10	1.5	4	10	11	6	2	3	33
Shields, 1995	1287	44	5	15	0.2	3	1	1	1	1	0	3
COMS, 1997	188	44	5	16	1	3	10	8	3	2	1	23
Singh, 2006	240	11	1	15	1	3.5	2	2	1	0	0	5
Shields, 2009	2514	136	0.4	24	0.6	4.5	2	2	1	0	1	5
Shields, 2019	3806	72	0.1	20	0.1	6.7	0.7	0.4	0.4	0.2	0.2	2
Total (mean)	8606	478	2.44	14.83	0.82	4.19	6.52	5.04	2.60	0.69	1.47	19.00

Table 2: Study characteristics and design

Author, year	Design	Definition of growth	Method
Mims, 1977	Retrospective	No numerical definition	Exam, photo, ultrasound, and/or angiography (not all cases)
Char, 1977	Retrospective	No numerical definition	Exam, photo, ultrasound (not all cases)
Augsburger, 1989	Retrospective	No numerical definition	Exam, photo, ultrasound (not all cases)
Butler, 1994	Retrospective	No numerical definition	Exam, photo, ultrasound (not all cases)
Shields, 1995	Retrospective	0.5 mm in LBD, 0.3 mm in height	Exam, photo, ultrasound (all cases)
COMS, 1997	Prospective	Growth to medium or large	Exam, photo, ultrasound (all cases)
Singh, 2006	Retrospective	0.3 mm in any dimension	Exam, photo, ultrasound (all cases)
Shields, 2009	Retrospective	0.5 mm in height or LBD	Exam, photo, ultrasound (all cases)
Shields, 2019	Retrospective	0.5 mm in height or LBD	Exam, photo, ultrasound, OCT (all cases)

LBD=Longest basal diameter, OCT=Optical coherence tomography

Table 3: Tumor characteristics

Author, year	Thickness	LBD	Symptoms	Juxtapapillary location	Orange pigmentation	Subretinal fluid	Overlying drusen	RPE atrophy
Mims, 1977	+	+	•	•	+	+	•	•
Char, 1977	+	+	+	•	•	•		
Augsburger, 1989	+	+	+	+	+	+	0	0
Butler, 1994	+	+	+	0	+	+	0	0
Shields, 1995	+	+	+	+	+	+	+	0
COMS, 1997	+	+	•	+	+	+	0	0
Singh, 2006	+	+	+	+	+	+	0	0
Shields, 2009	+	+	+	+	+	+		
Shields, 2019	+	+	+	+	+	+		

+ = Positively associated, + = Not studied, 0 = Not associated, - = Negatively associated, LBD = Longest basal diameter, RPE = Retina pigment epithelium

Comparison with Small Choroidal Melanoma

In a small dataset of 42 eyes that were observed before intervention, the median time to detect growth was 9 months (range: 3–26 months).^[7] The mean growth rate in basal dimension was 1.8 mm/12 months (range, 0.0–7.4 mm [95% confidence interval: 1.32–2.28]). Among these biopsies proven small choroidal melanoma, the proportion of tumors demonstrating growth by the 1st year and 2nd year was 86% and 98%, respectively [Figure 3]. We see that the trend for the small melanoma data starts with a higher cumulative percent of growth, over 80%, that reaches 98% by year 2. For all the other studies, they start with a lower cumulative proportion of growth, below 40%, that reaches 64% by year 2 and 95% by year 5.

Discussion

It has been noted since the time of Gass^[3] and shown subsequently in a prospective study of COMS^[2] that tumors labeled as small choroidal melanoma are heterogeneous in nature representing a mixture of nevus and melanoma with varying proportions based on the inclusion criteria of the study.^[29] Therefore, there has been a move toward using broad descriptive terms such as IMT until better methods of identifying small choroidal melanoma can be established.^[4] Even with the use of machine learning, the diagnostic predictive

value of diagnosing small choroidal melanoma does not go beyond 85%.^[19]

Time-to-event analysis of tumors led to the identification of two subsets of tumors, those that show observable growth in the first 2 years (early growth) and those that were observed to be growing over subsequent years (late growth). Over the years, the criteria of growth have become rigorous and supported by objective documentation with the use of ancillary imaging studies. Those with the observed growth in the first 2 years represent the most growing tumors (range 52%–83%, mean 67%) in every identified published study (9 studies, 8606 tumors) included in this review. The range of 52%–83% represents selection bias as all of the studies except one (COMS)^[2] were retrospective and some selected small tumors were treated and the growth rate, therefore, is only reported for cases that were selected to be observed. Of note, in the only prospective study (COMS), the 74% were observed to grow in the first 2 years.^[2]

If the documentation of growth is to be believed to be representing the malignant transformation of choroidal nevus, then the time-to-event analysis would reveal about equal proportion of benign tumors (nevus) growing (transforming to melanoma) over time. Skewed distribution with disproportionate greater number in the first 2 years (early growth, 67%) compared with those undergoing late growth (37%) suggests these that these two sets of the tumors are not the same. With

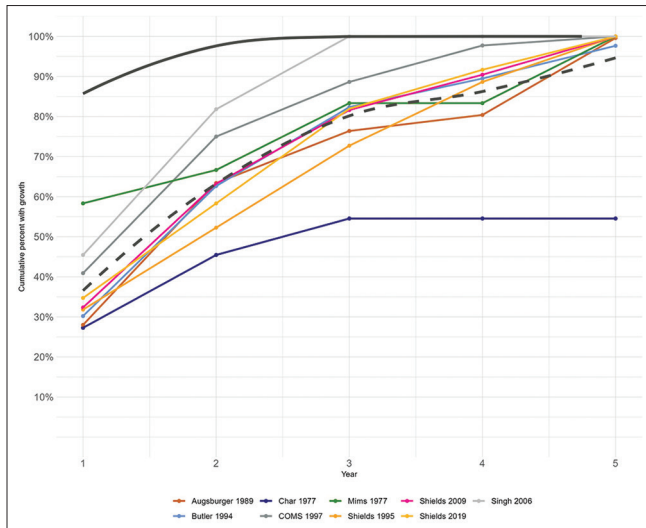


Figure 3: Biopsy-proven melanomas demonstrated growth at the 1st year and 2nd year in 86% and 98%, respectively, from a separate dataset (solid black) (Raval et al: REF 7). Studies collected from the literature start with a lower cumulative proportion of growth, below 40%, and reach 64% by year 2 and 95% by year 5 (average, dashed black)

recent publication of growth rate of biopsy-proven small choroidal melanoma, it becomes apparent that those tumors that were observed to show early growth were already malignant (melanoma) because their growth (mean increase in LBD of 1.8 mm/year (range 0.4–3.4) would be observable within 1–2 years with currently available imaging studies.^[7] We suggest that such tumors would be better characterized as “melanoma in evolution” rather than growing nevus or malignant transformation of the nevus. Consequently, therapeutic intervention rather than continued observation would be advisable. As a corollary, the so-called risk factors of growth^[32] could be considered diagnostic features of small choroidal melanoma. In fact, in a recently reported predictive model of small choroidal melanoma that had a high but not perfect discrimination score (0.86), the top five features predictive of melanoma diagnosis (such as presence of orange pigment, subretinal fluid, greater height, proximity to disc, and absence of drusen) were identical to the commonly reported risk factors.^[19]

Segregation of IMT with early growth from those with late growth can also bring clarity about the clinical nature of the latter. In contrast to choroidal melanoma growth, choroidal nevi enlarge at a much slower average rate of 0.06 mm/year, which will not be observable using the currently used imaging technique over the period of 2 years.^[27] Documented stability of the tumor during the initial 2 years of observation would argue against a diagnosis of malignancy and support continued observation rather than early intervention. The COMS small tumor study described a 1% melanoma-specific mortality rate in 5 years from those who were not immediately treated.^[34] A careful limited period of

observation undertaken as active surveillance was not found to increase the risk of metastasis.^[14,21–23] Since the risk of growth is higher in the initial 2 years and less after that, the interval of observation can be adjusted to reflect this temporal profile. The author’s preference is to recommend an initial review in 3 months, then at 6 months (so as not to overlook melanoma in evolution), and if stable at these 2 initial visits, an annual review with imaging studies is recommended. The fact that a proportion of tumors showing late growth were consistently below 15% across all studies with defined growth criteria and those using ancillary studies to assess for growth seems to suggest that this subset of tumors is true representative of malignant transformation of the IMT.^[34] Future studies on risk factors of growth could compare risk features between the “early growth” group and the “late growth” group. Studies on risk factors of malignant transformation should only include those cases that have already documented stability for 2 years.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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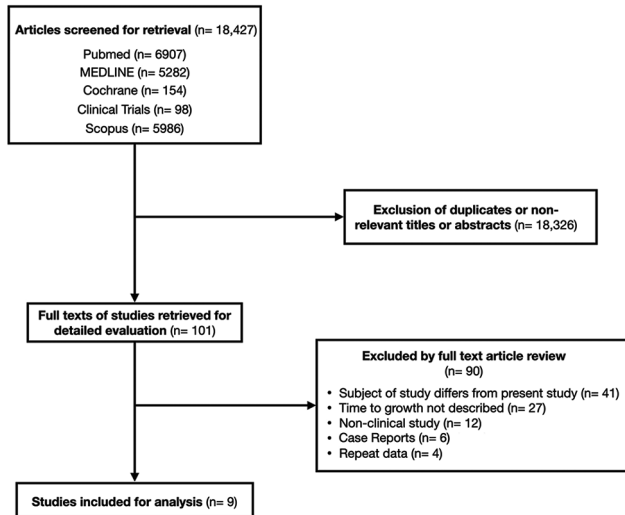
Conflicts of interest

Arun D. Singh reported having relevant financial activities outside the submitted work: Aura Biosciences (consultancy), IsoAid LLC (consultancy), Immunocore (consultancy), and Isoaid (consultancy). No other disclosures were reported. Zackery Oakey, Emily Zabor, and Yağmur Seda Yeşiltaş have no conflicts of interest to report.

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Supplement Figure 1: Flow diagram outlining literature search